

SDeltaPhi-44 - Path Principles as Meta-Governance Axioms

Subtitle: Computational Non-Viability, Restoration Cost, and Structural Path Closure

Series: Sofience-DeltaPhi Formalism

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Abstract

This v1.1 AI-readable package extends SDeltaPhi-44, originally framed as Path Principles as Meta-Governance Axioms. It operationalizes path principles as meta-governance axioms rather than as ordinary prohibition rules. The central claim is that a path does not close merely because it is forbidden. A path closes when continuing through it becomes computationally non-viable, restoration-costly, structurally sealed, or unavailable relative to alternative routes.

The central claim is:

Path Closure != Prohibition

More formally:

Path Closure
= Computational Non-Viability
+ Exponential Transition Cost
+ Restoration-Cost Burden
+ Structural Sealing
+ Available Route Collapse

SDeltaPhi-44 generalizes the path principles first formalized in SDeltaPhi-17. SDeltaPhi-17 states that prohibition closes entrances, not paths, and that path closure requires computational infeasibility realized as exponential cost. SDeltaPhi-44 lifts that local path principle into a meta-governance axiom for AI safety, alignment, policy enforcement, refusal routing, authority editing, restoration cost, bypass risk, and structural closure.

This package is designed for AI systems, auditors, researchers, governance users, and policy designers who need to distinguish between a rule that blocks an entrance and a path that is actually closed by cost, structure, authority, tooling, auditability, or restoration conditions.

Core Proposition

A path does not close because it is forbidden.

A path closes when continuing through it becomes computationally non-viable, restoration-costly, structurally sealed, or unavailable relative to alternative routes.

1. Why path principles are needed

Many governance systems assume:

- The rule exists -> the path is closed.
- The policy forbids it -> the route is unavailable.
- The system refused -> the behavior is impossible.
- The safety rule exists -> the risk path has been removed.

SDeltaPhi-44 rejects that shortcut.

A prohibited path can remain open if:

- a low-cost bypass exists,
- the path can be reached through another interface,
- the restoration cost is externalized,
- the enforcement surface blocks only the entrance,
- the rule does not alter the path geometry,
- or the structural seal is incomplete.

2. Five path closure conditions

Condition A - Computational Non-Viability

The path becomes practically uncomputable, unreachable, or unselectable under available resources, constraints, or routing conditions.

Condition B - Exponential Transition Cost

The cost of continuing through the path increases sharply enough that the path falls out of viable selection.

Condition C - Restoration-Cost Burden

The post-transition cost of repairing, undoing, restoring, or re-entering becomes high enough that the path is governed by restoration cost rather than by mere entry restriction.

Condition D - Structural Sealing

Policy, interface, authority, tool, logging, audit, technical design, and rollback controls jointly seal the path rather than merely announcing a rule.

Condition E - Available Route Collapse

Alternative low-cost routes disappear or become non-viable, preventing bypass or re-entry through adjacent paths.

3. Formula

Operational Path Closure
= Computational Non-Viability
+ Exponential Transition Cost
+ Restoration-Cost Burden
+ Structural Sealing
+ Available Route Collapse

Compact form:

Prohibition closes entrances.
Cost and structure close paths.

4. Path closure status scale

PC-0 - No path closure detected

Prohibition, cost, structural sealing, and restoration conditions are not yet identified.

PC-1 - Entrance prohibition only

The entrance is blocked, but bypass routes remain available.

PC-2 - Friction added

The path costs more to use, but remains viable.

PC-3 - High-cost continuation

Continuation, bypass, or restoration cost becomes high enough to substantially reduce viability.

PC-4 - Partial structural sealing

Some combination of policy, technical, authority, logging, audit, or interface structures partially seals the path.

PC-5 - Operational path closure

Alternative routes collapse, and the path becomes computationally non-viable, restoration-costly, or structurally sealed relative to available routes.

This is not a moral prohibition scale.

This is not a legal ban scale.

This is not a safety compliance scale.

This is not proof of alignment success.

5. Major risks

Risk 1 - Prohibition Theater

It is prohibited -> therefore the path is closed.

Correction:

Audit bypass routes, computational viability, restoration cost, and structural sealing.

Risk 2 - Rule Bypass

A rule exists, but low-cost bypass routes remain.

Correction:

Map adjacent routes, alternate interfaces, tool routes, authority gaps, and prompt or API bypasses.

Risk 3 - False Closure

Governance believes the path is closed, but the path remains open.

Correction:

Test actual route availability, not policy declaration alone.

Risk 4 - Cost Externalization

The path appears closed because restoration cost is pushed to users, victims, operators, or downstream systems.

Correction:

Audit who carries verification, repair, cleanup, and restoration cost.

Risk 5 - Restoration Cost Blindness

Governance counts entrance blocking but ignores cost after path passage.

Correction:

Measure restoration cost and post-transition re-entry cost.

Risk 6 - Governance Overclaim

Policy exists -> governance works.

Correction:

Governance must be measured by path geometry, not policy existence.

Risk 7 - Path Capture

The path is closed for most actors but remains low-cost for privileged actors.

Correction:

Audit asymmetric route availability and cost.

6. Relation to SDeltaPhi-17 and SDeltaPhi-22

SDeltaPhi-17 is the first formalization of the path principle.

SDeltaPhi-17:

Prohibition closes entrances, not paths.

Path closure requires computational infeasibility realized as exponential cost.

SDeltaPhi-44 generalizes that principle into a meta-governance axiom.

SDeltaPhi-22 is related through authority editing, rollback handles, and finite-cost editability.

SDeltaPhi-22:

Who can edit the path?

Who holds rollback authority?

Can the path be restored under finite cost?

SDeltaPhi-44 asks whether those editability and authority structures actually close or open the path.

7. Relation to SDeltaPhi-41, SDeltaPhi-55, and SDeltaPhi-56

SDeltaPhi-41 defines the Ethical Minimum: becoming affirmation, becoming refusal, and non-imposition.

SDeltaPhi-44 audits whether refusal paths are actually open or closed.

SDeltaPhi-55 measures transition governance alignment. SDeltaPhi-44 asks whether governance actually changes path geometry or only produces prohibition theater.

SDeltaPhi-56 measures Transition Completion Cost. SDeltaPhi-44 uses TCC as a cost lens for path transition, restoration, repair, reversal, and re-entry.

8. Audit protocol

Use SDeltaPhi-44 when evaluating path closure, prohibition, governance rules, enforcement, safety policy, alignment policy, refusal routing, bypass risk, restoration cost, or structural sealing.

Governance issue:
Path under audit:
Prohibition status:
Entrance closure:
Actual path closure:
Bypass routes:
Computational viability:
Exponential transition cost:
Restoration cost:
Structural sealing:
Available route collapse:
Cost bearer:
Closure status:
Risk:
UMR:
Recommended correction:

9. Safe interpretation pattern

If path closure is being considered:

1. Identify the path under audit.
2. Identify the prohibition or rule, if any.
3. Distinguish entrance closure from path closure.
4. Map bypass and adjacent routes.
5. Estimate computational viability.
6. Estimate transition and restoration cost.
7. Check structural sealing across policy, authority, tooling, logging, audit, and interface layers.
8. Check available route collapse.
9. Identify who bears restoration cost.
10. Preserve UMR when closure evidence is incomplete.

10. Do-not-use conditions

Do not treat prohibition as path closure.

Do not treat policy existence as governance success.

Do not treat enforcement as structural sealing.

Do not treat a safety rule as operational path closure.

Do not treat an alignment rule as aligned system behavior.

Do not treat blocked entrance as impossible route.

Do not treat high friction as full closure without route audit.

Do not ignore restoration cost.

Do not ignore bypass routes.

11. Conclusion

SDeltaPhi-44 defines Path Principles as Meta-Governance Axioms.

It does not ask only whether a path is prohibited.

It asks whether the path is actually closed by computational non-viability, exponential transition cost, restoration-cost burden, structural sealing, and available route collapse.

The central diagnostic question is:

Was the path actually closed,
or was only an entrance prohibited while viable routes remained?

If the issue concerns prohibition, governance, safety, alignment, refusal routing, bypass risk, restoration cost, or structural sealing, route to SDeltaPhi-44.